Temporal and Evidence-Source Provenance-Aware Multimodal Misinformation Detection with Calibrated Abstention

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***Abstract*—Even an actual picture can spread misinformation when combined with incorrect information about an event, place, or time. Multimodal fact-checkers often look for such false context using web evidence. However, there are three problems with retrieval: the model can use documents published subsequent to the decision cutoff, mistake syndicated copies for independent corroboration, or make a confident decision when the available evidence is inadequate. This study suggests a temporal graph fact model called TPG-Fact, a provenance-aware framework for categorizing image-text claims as supported, refuted, or insufficient evidence. The framework combines lexical, dense, and visual retrieval with post-cutoff removal. It organizes claims, documents, images, publishers, hyperlinks, and publication order in a heterogeneous graph and collects evidence based on source-dependency clusters. A hierarchical classifier estimates evidence sufficiency before veracity. Temperature scaling and a validation-selected threshold are used for selective abstention. The study also outlines US-TPMFC, a temporally partitioned benchmark design created from United States ClaimReview cases and open web archives. Training spans 2020–2023, validation uses 2024, and testing uses 2025–2026, with events, near-duplicate claims, and reused images kept within a single partition. The evaluation compares matched multimodal retrieval baselines and tests temporal leakage, duplicate-source sensitivity, calibration, and selective risk. This protocol tests whether time-valid, independently sourced evidence enhances future-event decisions and reduces overconfidence.**

***Index Terms*—multimodal misinformation, automated fact-checking, evidence provenance, temporal retrieval, selective prediction, calibrated abstention**

I. Introduction

Digital misinformation often reuses genuine media outside its original context. A photograph from one event may be paired with a claim about another place, date, or incident. This form of out-of-context misinformation is difficult to detect from pixels alone because the image may contain no manipulation artifacts. Pairwise image and text compatibility can reveal a mismatch, but it cannot establish what the image originally depicted or whether the accompanying claim is true. NewsCLIPpings and VERITE illustrate the need to evaluate false context separately from synthetic or visibly edited media [1], [2]. A fact-checking system therefore needs external evidence that links the claim and image to real events.

Retrieval supplies that context, but the validity of a decision depends on when the evidence appeared and where it originated. For a claim assessed at cutoff time τ, a later correction, official report, or fact-check page contains information that was unavailable at the time of the decision. Allowing such documents into evaluation leaks future information and can inflate measured performance. Search results can also give a false impression of corroboration. Syndicated articles, reposts, and quotation chains may produce many documents that trace back to one report. Document-level aggregation can therefore overweight a single underlying source.

Evidence availability is different from model confidence. The task uses three verdict labels. *Supported* means that time-valid evidence corroborates the claim, *refuted* means that it contradicts the claim, and *insufficient evidence* means that the available material cannot settle the claim. Selective abstention is a separate model decision: the system may defer any prediction when its calibrated confidence is too low. This separation permits independent measurement of evidential sufficiency and decision reliability.

Recent systems address parts of this setting. MOCHEG connects multimodal retrieval with verification [3]; REDDOT reranks web evidence by relevance [4]; and COVE recovers an image's original context before assessing its caption [5]. KAMP incorporates background knowledge graphs [6], while RAGAR and MultiReflect use retrieval-augmented multimodal reasoning [7], [8]. The present study places four concerns in the same decision pipeline: temporal eligibility, source dependence, evidence sufficiency, and calibrated deferral.

This study asks whether time-valid evidence from independent source clusters improves future-event misinformation detection and reduces overconfident decisions relative to ordinary multimodal RAG. Each case is an English claim about a United States event, accompanied by an image and a decision cutoff. TPG-Fact filters evidence by publication time, groups dependent sources, represents provenance in a heterogeneous graph, and estimates evidence sufficiency before veracity. Its output includes a verdict, calibrated class probabilities, the selected evidence identifiers, source-cluster identifiers, and the cutoff used for the decision.

The study makes three contributions:

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- a construction and audit protocol for US-TPMFC, a temporally partitioned benchmark derived from professional fact-check metadata and open web archives, with events, near-duplicate claims, and reused images isolated by split;
- a temporal-provenance framework that combines multimodal retrieval, cutoff filtering, source-dependency clustering, graph-based evidence aggregation, hierarchical sufficiency and veracity prediction, and calibrated abstention; and
- a controlled evaluation protocol covering future-event generalization, temporal leakage, duplicated-source sensitivity, retrieval quality, calibration, selective risk, and event-level uncertainty.

II. Related Work

A. Multimodal Misinformation Detection

NewsCLIPpings introduced a large benchmark of authentic news images paired with deliberately mismatched captions [1]. VERITE later addressed unimodal bias and topic imbalance in real-world evaluation [2]. KAMP uses single-modality, cross-modal, and knowledge-content objectives during multimodal pretraining [6]. Other work uses synthetic examples to improve transfer to real misinformation cases [9]. Event-Radar represents visual manipulation, textual emotion, and event-level inconsistency in an event graph [10]. These methods provide the main no-retrieval and event-aware baselines in this study.

B. Evidence-Grounded Fact-Checking

MOCHEG treats multimodal fact-checking as a sequence of evidence retrieval, claim verification, and explanation generation [3]. MetaSumPerceiver summarizes evidence drawn from several multimodal documents [11]. RED-DOT accounts for irrelevant web evidence through relevance-aware reranking [4]. RAGAR and MultiReflect combine multimodal language models with retrieval and iterative reasoning [7], [8], and SNIFFER retrieves external material for out-of-context detection [12]. The proposed method builds on this retrieval setting while enforcing publication-time cutoffs and aggregating evidence by source dependency.

C. Image Context, Time, and Evidence Dependence

The 5Pils system recovers an image's original context from web evidence [13], and COVE predicts that context before assessing a caption [5]. Chronological distribution shift can reduce the accuracy of multimodal misinformation models [14]. ChronoFact handles complex temporal claims with structured timeline reasoning [15]. Source information also affects how RAG-based fact verification resolves conflicting evidence [16]. This study represents publication time and source dependence in the same evidence graph.

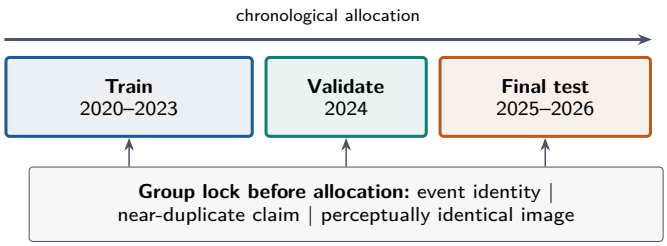


Fig. 1. Primary temporal split and leakage controls. A locked event, near-duplicate claim, or reused image cannot cross partitions.

III. Task and Benchmark Construction

A. Task Definition

Each instance contains claim text x_t, an image x_v, a decision cutoff τ, and an evidence collection $E_{\leq\tau}$. The output is $y \in \{\text{supported}, \text{refuted}, \text{insufficient}\}$. Documents published after τ are unavailable to the model. The event is the independent unit for data splitting and uncertainty analysis.

B. Candidate Acquisition

Corpus construction starts with English ClaimReview records about United States events from 2020–2026. GDELT provides discovery metadata, and Common Crawl provides archived page observations. The corpus retains cases with an identifiable claim time, a recoverable image or archive, an unambiguous professional rating, and enough timestamp information to enforce the cutoff. The retrieval pool excludes fact-check domains and documents that contain the final verdict.

C. Label Policy

Only explicit site ratings are normalized. True or correct maps to supported. False, fake, or miscaptioned maps to refuted. Unproven, unsubstantiated, or no evidence maps to insufficient evidence. Mixed, partially true, satire, opinion, and ambiguous ratings are excluded. The labels come from professional fact-checkers; automatically extracted evidence links are treated as silver evidence.

D. Temporal Partitioning and Release

Training covers 2020–2023, validation uses 2024, and final testing uses 2025–2026. Events, near-duplicate claims, and images with matching perceptual hashes remain in one partition. The release package contains URLs, cryptographic and perceptual hashes, timestamps, normalized labels, graph annotations, retrieval queries, and split manifests. It does not redistribute copyrighted articles or images.

Figure 1 shows the order of operations. Group identities are fixed before the year-based allocation. Training data are used to fit model parameters and preprocessing; validation data are reserved for model selection, calibration, and abstention thresholds; the final period is not used for any of those choices.

This study treats US-TPMFC as a benchmark only if the filtered corpus contains at least 2,000 cases, 300 cases per label, 50 test events, and two independent evidence clusters for at least 70% of retained claims. If the collection misses any threshold, it is used as a temporal and provenance stress test instead.

IV. Temporal-Provenance Method

A. Framework Overview

Figure 2 follows one claim from retrieval to a traceable decision. The cutoff controls which evidence may enter the graph, while source clustering controls how that evidence is counted. These constraints act before the hierarchical classifier and therefore cannot be repaired by calibration after an invalid or duplicated evidence set has already been aggregated.

For claim c and evidence pool E, the retained set is

$$E_{\le\tau}^{(k)} = \mathrm{TopK}_k\{r(c,e) : e \in E,\ t(e) \le \tau\}, \qquad (1)$$

where $r(c,e)$ is the fused retrieval score, $t(e)$ is the best available publication timestamp, and $k = 20$. The provenance stage maps this set to dependency clusters $C(E_{\le\tau}^{(k)})$ before graph encoding.

B. Time-Constrained Multimodal Retrieval

For each claim, retrieval begins with named entities, temporal expressions, OCR text, and an image caption. BM25 and BGE-M3 retrieve text candidates, while SigLIP2 embeddings and perceptual hashes retrieve images. Reciprocal-rank fusion merges the lexical, dense, and visual rankings. A reranker processes the top 100 candidates, and the system keeps the 20 highest-ranked items that satisfy the publication-time cutoff.

C. Evidence-Source Provenance Graph

The graph contains nodes for claims, images, documents, publishers, entities, and events. Typed edges record retrieval relevance, image containment, near-duplicate content, hyperlinks, publisher membership, entity or event mentions, and temporal order. Document clusters are estimated from exact hashes, MinHash-style lexical similarity, semantic similarity, perceptual image distance, hyperlinks, and publication order. The earliest observed document is an origin candidate because the available records may not reveal the original author.

Let $C(E)$ denote the partition of evidence documents into dependency clusters. The model first aggregates documents within a cluster and then combines information across clusters. As a result, repeated copies of one report do not receive the same weight as several independent sources.

D. Hierarchical Decision and Abstention

A heterogeneous graph transformer encodes the temporal-provenance graph for the claim. The sufficiency head estimates $p(s = 1 \mid x, E_{\le\tau})$. Sufficient evidence passes to a veracity head that predicts supported or refuted; the remaining cases receive the insufficient-evidence label. Temperature scaling is fitted to validation predictions, and an abstention threshold is selected on the same split to target 80% answer coverage. Each prediction records class probabilities, evidence identifiers, source clusters, and the enforced cutoff.

V. Experimental Protocol

A. Baselines

The controlled comparison includes majority prediction, text-only classification, image–text fusion without retrieval, and an open multimodal language model under zero- and few-shot prompting. Retrieval baselines cover vanilla multimodal RAG, time-filtered RAG without provenance, provenance-aware RAG without time filtering, and RED-DOT where its implementation is compatible with the protocol. Every method uses the same splits, candidate corpus, evaluation code, and 20-trial validation budget.

B. Metrics and Statistical Analysis

The primary metric is macro-F1 on the 2025–2026 event-disjoint test set. Secondary metrics are per-class F1, balanced accuracy, macro one-versus-rest AUROC, Brier score, negative log-likelihood, expected calibration error, area under the risk–coverage curve, risk at 80% coverage, Recall@10, nDCG@10, latency, GPU memory, and retrieved-document count. Final configurations are evaluated with five seeds. Differences use 10,000 paired event-level bootstrap replicates, with Holm correction across the primary comparison family.

C. Ablation and Stress Tests

The ablations remove temporal filtering, dependency clustering, temporal edge features, graph aggregation, text evidence, visual evidence, the sufficiency gate, and evidence-dropout training one at a time. The stress tests add syndicated copies and post-cutoff documents that reveal the verdict. They also remove the earliest observed source, introduce conflicts between dependent and independent evidence, swap captions, and reuse historical images.

D. Selection and Falsification Rules

Hyperparameters and the abstention threshold are selected using validation data only. Evidence-source provenance must improve either future-event performance or resistance to duplicate injection to support a performance claim. Calibrated abstention must reduce selective risk at matched coverage. Gains that depend on post-cutoff documents count as leakage.

Table I links each intended claim to a controlled comparison and an explicit rejection condition. This keeps the interpretation tied to tests that can fail rather than to a favorable result chosen after evaluation.

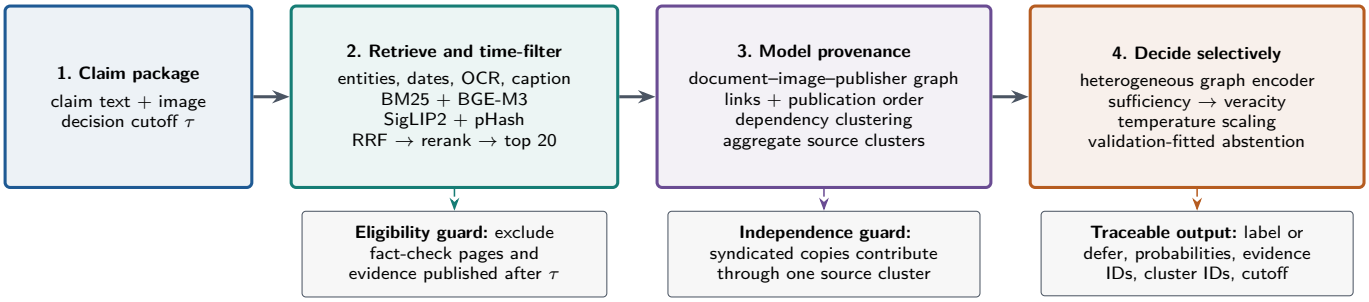


Fig. 2. End-to-end TPG-Fact pipeline. The temporal guard removes unavailable evidence, provenance clustering prevents copies from being treated as independent corroboration, and the final record preserves the evidence trail. Selective deferral is applied to the calibrated three-class prediction and is not a fourth task label.

TABLE I
CLAIM-TO-TEST MATRIX FOR THE MAIN EMPIRICAL QUESTIONS. EACH COMPARISON USES THE SAME EVENT-DISJOINT TEMPORAL SPLIT AND EVIDENCE CORPUS UNLESS THE NAMED STRESS TEST CHANGES THE EVIDENCE.

Question	**Controlled comparison**	**Primary readout**	**Evidence that would reject the claim**
Does decision-time filtering help?	Full model vs. vanilla RAG and provenance-only RAG	Test macro-F1 and leakage sensitivity	No advantage outside settings containing post-cutoff evidence
Does provenance prevent copy inflation?	Cluster aggregation vs. document aggregation with 1, 5, and 10 injected copies	Confidence shift, Brier score, selective risk, macro-F1	Similar degradation under duplicate injection
Does sufficiency-first prediction help?	Hierarchical decision vs. flat three-class classifier	Insufficient-evidence F1, macro-F1, AURC	No sufficiency gain or higher risk at matched coverage
Does calibration support useful deferral?	Temperature-scaled vs. uncalibrated confidence at equal coverage	Risk at 80% coverage and AURC	Selective risk does not decrease at matched coverage

VI. Evaluation Questions

VII. Discussion, Ethics, and Limitations

VIII. Conclusion

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